IN104: Monte Carlo Simulation of the Ising Model

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March 2026

1 Introduction

Monte Carlo simulations are a powerful class of numerical methods used to study complex systems across many fields, including physics, biology, chemistry, quantitative finance, and applied mathematics. The central idea is to rely on randomness to explore a probability distribution and estimate quantities of interest (observables), together with their statistical uncertainties.

In this project, we apply this methodology to a simple yet fundamental model from statistical physics: the Ising model. Despite its simplicity, this model captures the essential mechanisms behind ferromagnetism and provides an ideal framework for learning modern computational techniques.

1.1 Project Overview

Some materials exhibit a ferromagnetic phase, meaning they develop a spontaneous and persistent magnetization due to the collective alignment of microscopic magnetic moments. However, this ordered behavior only exists below a certain temperature, called the critical temperature T_c. Above this temperature, thermal fluctuations destroy the alignment, and the material becomes disordered (paramagnetic phase).

To model this behavior, we introduce a simplified description:

- Each atom is represented by a *spin*
- Each spin can take only two values:

$$S_i = +1 \quad (\text{“up”}) \qquad \text{or} \qquad S_i = -1 \quad (\text{“down”}) \tag{1}$$

1.1.1 The Ising Model

The energy of a configuration of spins $\{S_i\}$ is defined as:

$$E(\{S_i\}) = -J \sum_{\langle m,n \rangle} S_m S_n \tag{2}$$

where J is the interaction strength and $\langle m, n \rangle$ denotes pairs of nearest neighbors.

When $J > 0$, the energy is minimized when neighboring spins are aligned. At zero temperature ($T = 0$), the system reaches its ground state, in which all spins are aligned (either all $+1$ or all -1). These two configurations are equivalent and correspond to the ferromagnetic phase.

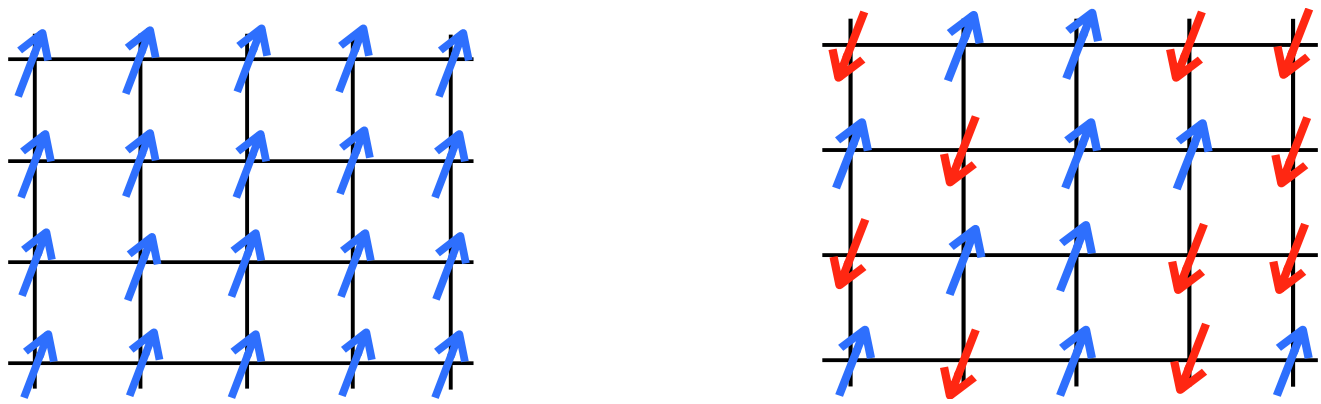
1.1.2 Role of Temperature

At finite temperature, thermal fluctuations introduce disorder, and the system explores multiple configurations. The probability of observing a configuration m, denoted $\{S_i^m\}$, is given by the Boltzmann distribution:

$$P(\{S_i^m\}) = \frac{1}{Z} e^{-E(\{S_i^m\})/T} \tag{3}$$

where Z is the partition function, a normalization constant ensuring that:

$$\sum_m P(\{S_i^m\}) = 1 \tag{4}$$



(a) $T = 0$: ferromagnetic phase, $\langle m \rangle = 1$

(b) High T: paramagnetic phase, $\langle m \rangle \approx 0$

Figure 1: Comparison between ordered (ferromagnetic) and disordered (paramagnetic) phases.

1.2 Methods and Objectives

The goal of this project is to use Monte Carlo methods to sample configurations of the Ising model according to the Boltzmann distribution. To achieve this, we will implement the Metropolis algorithm.

1.2.1 Metropolis Algorithm

The Metropolis algorithm provides an efficient way to sample the Boltzmann distribution. A brute-force approach would require evaluating all possible configurations, which is computationally infeasible: for example, a 10×10 lattice contains 100 spins, corresponding to 2^{100} possible configurations.

The algorithm proceeds as follows:

- (1) Start from an initial configuration $\{S_i^{(0)}\}$
- (2) Compute its energy E_0
- (3) Select a spin at random and propose to flip it
- (4) Compute the new energy E_n
- (5) Accept the move:
  - always if $E_n < E_0$
  - otherwise with probability $e^{-(E_n - E_0)/T}$
- (6) Update the configuration accordingly
- (7) Measure observables and repeat the process (go back to step (3))

Importantly, there is no need to compute the partition function Z, since it cancels when taking ratios of probabilities. Also, It is of paramount importance that observables are computed even if the move have been rejected.

This procedure generates a Markov chain: each new configuration depends only on the current one. Under general conditions (ergodicity and detailed balance), the sampled configurations converge to the desired Boltzmann distribution.

1.3 Computing Observables

From the generated samples, we will compute macroscopic observables such as the magnetization and study their dependence on temperature. In particular, we will identify the phase transition and estimate the critical temperature T_c, which can be compared to the exact analytical result for the two-dimensional Ising model:

$$T_c \approx 2.269\, J \tag{5}$$

A second objective of the project is to assess the reliability of Monte Carlo simulations. We will implement post-processing tools to compute:

- mean values and statistical uncertainties
- autocorrelation functions
- binning analysis

These techniques are essential when dealing with correlated data, and allow for a proper estimation of error bars in Monte Carlo simulations.

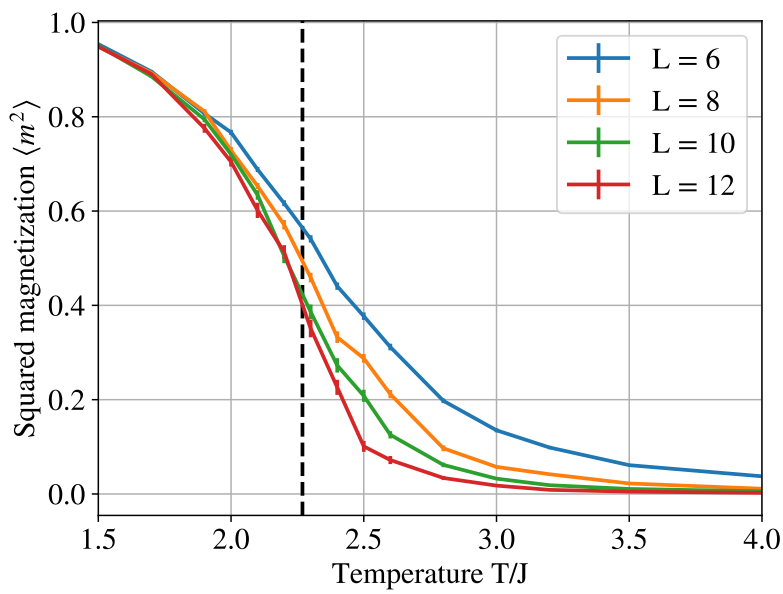


Figure 2: Squared magnetization vs. temperature

1.4 Postprocessing Tools

1.4.1 Observables

Each time we have a configuration, we can calculate the values of different observables. For example we compute the total energy associated to that specific configuration, which is also used to accept or reject the next move. A key quantity that we want to calculate is the average magnetization:

$$m = \sum_{m}^{N} S_m / N \tag{6}$$

where $N = L_x \times L_y$ is the system size. It is important to store the values of observables and save them to file, such that when the simulation is finished we can compute the mean value and the standard deviation.
It turns out that the average magnetization is not a good quantity to determine if we are in the paramagnetic or ferromagnetic phase: when averaging over all the samples, it may happen that we go from a ferromagnetic phase with negative magnetization to one with positive magnetization. To overcome this problem, we will also store the squared magnetization:

$$m^2 = (\sum_{m}^{N} S_m / N)^2 \tag{7}$$

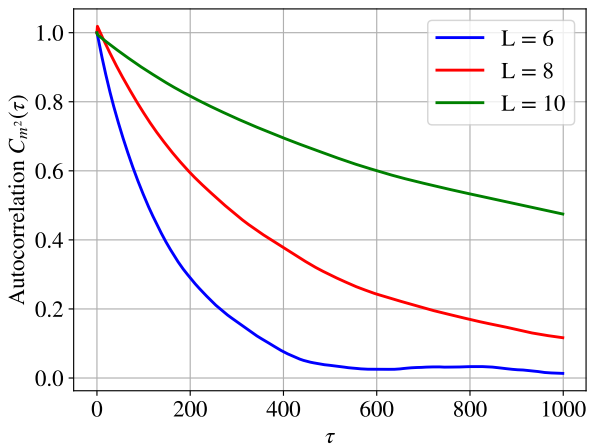
Note: the square must be taken for each sample and then we will average these values over all the samples. In Fig. 2, we can see how the squared magnetization changes when tuning temperature, leading to a phase transition from a ferromagnetic to paramagnetic phase.

1.4.2 Statistical Errors

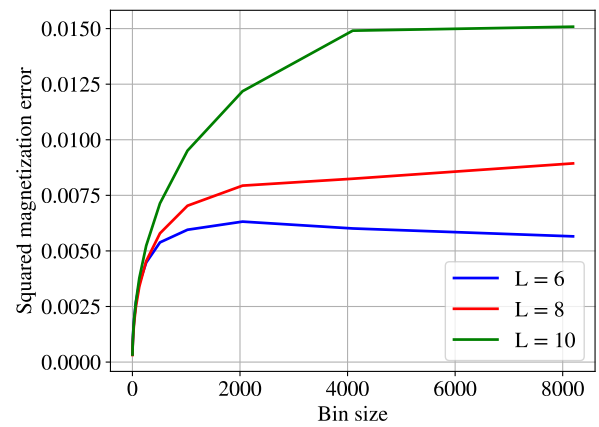
In simulations like Monte Carlo, each result must come with an uncertainty. It turns out that taking the standard deviation of all the values of a given observables O usually underestimate the error. The reason is that each configuration is strongly correlated to the previous ones. To quantify this, we compute the autocorrelation function:

$$C_O(\tau) = \frac{\langle O(t)O(t+\tau)\rangle - \langle O(t)\rangle\langle O(t)\rangle}{\sigma_O^2} \tag{8}$$

In Fig. 3a, a typical autocorrelation function in the Metropolis algorithm is shown.
To overcome this problem, we perform a binning analysis: we group data into bins of different sizes and calculate the mean value of each bin. We then calculate the standard deviation over all bins and plot it as a function of the bin sizes. We see that when the bin size matches with the autocorrelation time, the error saturates to the upper bound, which is the estimation of error. (see fig.3b).



(a) Autocorrelation at $T = 2.3J$, close to the phase transition.



(b) Binning analysis at $T = 2.3J$, close to the phase transition.

Figure 3: Post-processing of Monte Carlo $\langle m^2\rangle$ observable.